Questions, Structure, and Causal Simulation

Observable Quotients, Reachability, and the Information Required for Selection

Abstract

We formulate a sequence of constructions from response maps to exact controlled simulation. Static response profiles determine a quotient; closure under future experiments yields a canonical deterministic behavioral quotient. A finite stochastic analogue preserves all adaptive transcript laws and therefore preserves exact causal seed cost. We then separate universal reachability, invariance, random generation, and targeted selection. Universal reachability forces every transition invariant to be constant, whereas reliable selection among N prescribed targets requires a quantified information flow through the controls. Finally, empty inputs, unspecified inputs, and symmetry-invariant selection are distinguished by elementary impossibility results. The note is a self-contained synthesis of standard constructions with explicit connections to exact one-seed simulation; the results are not asserted to be new to the literature.

1 Scope and mathematical context

All collections of states and questions below are sets. Shannon entropy is measured in bits. No interpretation of a set as a physical substrate is assumed. The constructions concern representations and controlled transition laws, rather than an origin of mathematical axioms.

Behavioral equivalence and canonical quotients belong to automata and coalgebraic system theory [1]. Minimal predictive representations of stochastic processes are developed in computational mechanics [2]; input–output processes require an explicitly controlled formulation [3]. Stochastic aggregation by bisimulation is treated in [4]. Our deterministic quotient is a specialization of these standard ideas. Our stochastic result below assumes strong lumpability and does not identify it with every weaker notion of observational equivalence.

2 Questions define an observable quotient

Let $\Omega \neq \varnothing$. For each q in a set Q, let Y_q be a set and let $A_q : \Omega \to Y_q$ be a response map. Define

$$A_Q : \Omega \longrightarrow \prod_{q \in Q} Y_q, \qquad A_Q(x) = (A_q(x))_{q \in Q},$$

and $x \sim_Q x'$ if $A_Q(x) = A_Q(x')$.

Theorem 2.1 (Observable factorization)**.** *The map $[x]_Q \mapsto A_Q(x)$ is a bijection*

$$\Omega/{\sim_Q} \;\cong\; \operatorname{im} A_Q.$$

Every function $f : \Omega \to Z$ constant on $\sim_Q$-classes factors uniquely through the quotient projection. If $Q \subseteq Q'$, the map $[x]_{Q'} \mapsto [x]_Q$ is well defined and surjective. The response map A_Q is injective precisely when the questions separate all distinct points of Ω.

Proof. Two representatives have the same image exactly when they are equivalent. This proves well-definedness and injectivity; surjectivity onto the image is immediate. Define the factor of f by $\bar{f}([x]) = f(x)$. Constancy on classes proves well-definedness, and surjectivity of the projection proves uniqueness. Equality of all Q' responses implies equality of the Q responses, proving the refinement assertion. The last assertion is the definition of separation. □

For a specified response sequence $(q_j, y_j)_{j=1}^n$, the compatible set is

$$K_n = \bigcap_{j=1}^{n} A_{q_j}^{-1}(\{y_j\}).$$

Then $K_{n+1} \subseteq K_n$. This is an update of a description; the definition does not specify any state transition caused by asking a question. Such transitions must be supplied separately.

Remark 2.2. For $Q = \varnothing$, the empty product is a singleton. Thus a nonempty Ω has one observable class, not zero states. Absence of questions is different from an empty state space. Moreover, the bijection in Theorem 2.1 initially concerns sets; algebraic or topological structure requires additional hypotheses.

3 Future questions generate an exact simulator

Let $S \neq \varnothing$ be a state set, $A \neq \varnothing$ an action set, $o : S \to Y$ an observation map, and $F_a : S \to S$ a total transition for each $a \in A$. For a word $w = a_1 \cdots a_k \in A^*$, set $F_w = F_{a_k} \circ \cdots \circ F_{a_1}$ and $F_\epsilon = \mathrm{id}$. Define the response to a future experiment by $o_w = o \circ F_w$ and put

$$x \equiv x' \quad \Longleftrightarrow \quad o(F_w(x)) = o(F_w(x')) \quad \text{for every } w \in A^*.$$

Theorem 3.1 (Canonical deterministic behavioral quotient)**.** *The relation $\equiv$ is the largest equivalence relation R such that*

$$xRx' \Longrightarrow o(x) = o(x') \quad \textit{and} \quad F_a(x)RF_a(x') \quad (a \in A).$$

The quotient $M = S/\equiv$ has well-defined maps

$$\bar{o}([x]) = o(x), \qquad \bar{F}_a([x]) = [F_a(x)],$$

and reproduces every finite observation transcript under every deterministic observation-based adaptive policy.

If $e : S \to N$ is a surjection into another deterministic system with $o = g \circ e$ and $e \circ F_a = G_a \circ e$, there is a unique surjection $r : N \to M$ satisfying $r \circ e = [\cdot]$. It preserves observations and transitions. In particular, when finite, M has at most as many states as every such exact quotient N.

Proof. Equality of profiles gives an equivalence relation. The empty word ensures equality of present observations. For any w, $o(F_w(F_a(x))) = o(F_{aw}(x))$, so equivalent states have equivalent successors. Conversely, induction on word length shows that every observation-preserving transition congruence R relates only states with equal future profiles. Hence $R \subseteq \equiv$.

The quotient maps are well defined by these properties. Equal observation histories cause a deterministic policy to select the same next action. Induction then gives equal observations at every time. If $e(x) = e(x')$, the intertwining equations imply equality of all future observations. Thus $r(e(x)) = [x]$ is well defined. It is surjective and preserves the system maps; uniqueness follows from surjectivity of e. □

Proposition 3.2 (Finite refinement)**.** *Suppose $|S| = m < \infty$. Define R_0 by equality of o, and recursively*

$$xR_{k+1}x' \iff xR_kx' \text{ and } F_a(x)R_kF_a(x') \text{ for every } a.$$

Then R_k is equality of responses to words of length at most k. The sequence stabilizes at $\equiv$ after at most $m-1$ strict refinements.

Proof. The word-length assertion follows by induction. Each strict refinement increases the number of nonempty classes, which is at most m. Once two consecutive relations agree, the recursion shows they agree thereafter; the stabilized relation equals the intersection over all k. □

Example 3.3 (Present responses need not suffice)**.** Let $S = \{0, 1, 2\}$, $o(0) = o(1) = 0$, $o(2) = 1$, and let the sole action satisfy $F(0) = 0$, $F(1) = 2$, $F(2) = 2$. States 0 and 1 have equal present responses but different responses after one step. Their static identification cannot define an exact transition quotient.

4 Stochastic quotients and one seed

Let S, A, Y be finite nonempty sets and $K(s, a; y, s') \geq 0$ with $\sum_{y,s'} K(s, a; y, s') = 1$. The kernel generates output y and next hidden state s'. Let $e : S \twoheadrightarrow M$ be a surjection satisfying

$$e(s) = e(t) \Longrightarrow \sum_{s':e(s')=m'} K(s, a; y, s') = \sum_{s':e(s')=m'} K(t, a; y, s') \tag{1}$$

for every a, y, m'. Define the common quantity as $\bar{K}(e(s), a; y, m')$. The initial law on M is $e_*\nu$, where ν is the initial law on S.

Theorem 4.1 (Adaptive law preservation)**.** *The original and quotient systems have identical output-transcript laws under every deterministic policy based on past actions and outputs. Consequently their exact causal one-seed costs agree at every finite horizon. Their classes of admissible seed laws agree also at infinite horizon, so countable-seed existence, finite-entropy seed existence, and atomic capacity agree whenever defined through those laws.*

Proof. Inductively, the joint distribution of an observed history and the projected current state is the same in both models. Given a history, the policy selects the same action. Summing the next-step probabilities over the fibers of e and applying (1) proves the induction.

A causal seed is a random variable U and deterministic output functions of (U, h, a), required to reproduce all prescribed policy laws. The two models impose identical constraints on these same functions and the law of U. Thus their feasible seed laws, and any infimum or supremum depending only on those laws, coincide. At infinite horizon the equality concerns every finite cylinder and hence every policy trajectory law on the finite-alphabet product space. □

Proposition 4.2 (Finite causal seed existence)**.** *For any finite-horizon controlled output tree there is one finite seed exact under every deterministic adaptive policy.*

Proof. For each history–action pair (h, a) draw an independent variable $Z_{h,a}$ with the prescribed conditional output law at that node. Let U be the finite tuple of all these variables and let the decoder return $Z_{h,a}$. A reached history depends only on variables at earlier depths, so the current variable has the required conditional law. Induction proves correctness for every policy. A shared history–action pair always uses the same variable. This proves existence, not entropy optimality; it gives only $H(U) = \sum_{h,a} H(Z_{h,a})$. □

Remark 4.3 (Relation to the earlier seed model)**.** For a single action, the optimal finite-horizon cost is simply the entropy of the output prefix: deterministic data processing gives the lower bound, and the prefix itself is a seed attaining it. With multiple actions, one must satisfy all adaptive policies using one decoder. Theorem 4.1 preserves this stronger requirement and does not replace it with a maximum of individual transcript entropies.

5 Universal reachability constrains invariants

Write $x \rightsquigarrow y$ when $F_w(x) = y$ for some finite action word w.

Definition 5.1. The system is universally reachable if $x \rightsquigarrow y$ for every x, y. A transition invariant is a map $I : S \to V$ with $I(F_a(x)) = I(x)$ for all x, a.

Theorem 5.2 (Invariant obstruction)**.** *If $x \rightsquigarrow y$, every transition invariant satisfies $I(x) = I(y)$. Therefore universal reachability forces every transition invariant to be constant. More precisely, invariants are exactly the maps constant on each component of the undirected graph obtained by replacing every transition $x \to F_a(x)$ with an undirected edge.*

Proof. Iterating invariance along a word proves the first assertion. Equality along a directed edge also gives equality along the reversed undirected edge. Thus invariants are constant along every undirected path. Conversely, endpoints of a transition lie in the same such component, so every component-constant map is invariant. □

Remark 5.3. The converse of the second assertion is false. The system on $\{0, 1\}$ with $F(0) = 1$, $F(1) = 1$ has only constant transition invariants, but 1 cannot reach 0. Invariance under a smaller symmetry group is also different from invariance under all allowed dynamics.

Proposition 5.4 (Reachability descends to an exact quotient)**.** *If S is universally reachable, its behavioral quotient in Theorem 3.1 is universally reachable.*

Proof. For representatives x, y, choose w with $F_w(x) = y$ and project the path. The reverse implication need not hold: a constant observation can collapse disconnected hidden components to a single quotient state. □

6 Selection has an information requirement

The existence of a path to every target is different from selecting a requested target. Let Z be uniform on $\{1, \dots, N\}$, $N \geq 2$, and let U be a countably valued seed independent of Z. Let A denote a finite or countably valued control transcript, including any stopping decision, and suppose the final target estimate is $\widehat{Z} = f(U, A)$. The controls may depend on the requested target and on observations of the seeded system. Assume $H(A \mid U) < \infty$. Write $h_2(p) = -p \log_2 p - (1 - p) \log_2(1 - p)$, with $0 \log_2 0 = 0$.

Theorem 6.1 (Information required for target selection)**.** *If $\Pr(\widehat{Z} \neq Z) = \epsilon$, then*

$$H(A \mid U) \;\geq\; I(Z; A \mid U) \;\geq\; \log_2 N - h_2(\epsilon) - \epsilon \log_2(N - 1).$$

For exact selection, $H(A \mid U) \geq \log_2 N$. If there are at most K possible complete control transcripts for each seed value, exact selection requires $K \geq N$.

Proof. Set $E = \mathbf{1}_{\{Z \neq \widehat{Z}\}}$. Given $E = 0$ and $\widehat{Z}$, the target is known; given $E = 1$ and $\widehat{Z}$, at most $N - 1$ targets remain. The entropy chain rule gives

$$H(Z \mid \widehat{Z}) \leq h_2(\epsilon) + \epsilon \log_2(N - 1).$$

Since $\widehat{Z}$ is a function of (U, A), $H(Z \mid U, A) \le H(Z \mid \widehat{Z})$. Independence yields

$$I(Z; A \mid U) = H(Z \mid U) - H(Z \mid U, A) = \log_2 N - H(Z \mid U, A).$$

Finally $I(Z; A \mid U) \le H(A \mid U)$, and the support bound gives $H(A \mid U) \le \log_2 K$. □

Example 6.2 (Complete control with zero seed entropy)**.** Let $S = A = \{1, \dots, N\}$, $F_a(x) = a$, and $o(x) = x$. Every state reaches every target in one step. The simulator is deterministic, so its seed cost is zero. Choosing a uniform requested target means setting $A = Z$; the $\log_2 N$ bits occur in the controls, attaining Theorem 6.1.

Example 6.3 (Random generation without target control)**.** An actionless system outputs an independent uniform element of $\{1, \dots, N\}$ at each step. Every finite output word has positive probability, and its length-n seed cost is $n \log_2 N$. Yet at a fixed step the probability of matching an independent requested target is only $1/N$. Waiting until a match requires access to the target and a target-dependent stopping decision; those are additional controls.

7 Empty inputs and symmetry

In the category of sets, $\varnothing$ is initial and a singleton 1 is terminal [5].

Proposition 7.1 (Empty input versus unspecified input)**.** *For every set S there is exactly one map $\varnothing \to S$, and its image is empty. Maps $1 \to S$ correspond bijectively to elements of S. Thus a constant output from a singleton input is a specified choice of an element, whereas a map from the empty set produces no values.*

Proof. The graph of a map with empty domain is empty. A map from a singleton is uniquely determined by the image of its sole element. □

Theorem 7.2 (No permutation-invariant distinguished point)**.** *There is no assignment of a point $c(S) \in S$ to every nonempty set such that every bijection $f : S \to T$ satisfies $f(c(S)) = c(T)$.*

Proof. Take a two-element set S and the permutation interchanging its points. The asserted identity would make $c(S)$ a fixed point of that permutation, which has none. □

Corollary 7.3 (Random symmetry breaking has a seed cost)**.** *On an N-element set, the only probability law invariant under all permutations is uniform. Every discrete seed U deterministically generating a sample from that law satisfies $H(U) \ge \log_2 N$, and equality is attained. On a countably infinite set there is no countably additive probability law invariant under every permutation.*

Proof. Transpositions force all singleton probabilities to be equal. In the finite case their sum is one, so each equals $1/N$; entropy data processing gives the seed bound. In the countably infinite case a common positive mass has infinite sum, whereas common mass zero has sum zero. Neither gives a probability measure. □

Proposition 7.4 (Generation needs a starting set or a nullary rule)**.** *Let $R \subseteq S \times S$, $X_0 \subseteq S$, and $X_{n+1} = X_n \cup R[X_n]$. If $X_0 = \varnothing$, then $X_n = \varnothing$ for every n. If instead $X_{n+1} = X_n \cup R[X_n] \cup B$, new elements can appear from B; the set B is additional generating data.*

Proof. The image of the empty set under a relation is empty, so induction proves the first assertion. The second follows directly from the modified rule. □

8 Unobservable extensions

Theorem 8.1 (Hidden extensions are not identified by transcripts)**.** *Given a deterministic controlled system (S, F, o) and any nonempty set E, choose arbitrary functions $G_a : S \times E \to E$. On $S \times E$ set*

$$\widetilde{F}_a(s, e) = (F_a(s), G_a(s, e)), \qquad \widetilde{o}(s, e) = o(s).$$

For every initial (s, e) and every observation-based adaptive policy, the output transcript equals that of the original system started at s. Thus these transcripts cannot identify the choice of E or G_a.

Proof. Projection onto S intertwines transitions and observations. Induction on the transcript length proves the assertion, including adaptive choices. □

9 Consequences for exact simulation

The results give the following logically distinct steps:

$$\text{response maps} \longrightarrow \text{observable quotient} \longrightarrow \text{transition-compatible model.}$$

The second arrow requires closure under future observations or a lumpability condition. It does not follow from present responses alone. An exact behavioral reduction preserves the prescribed policy laws, and therefore their one-seed optimization problem. It neither implies universal reachability nor identifies unobserved extensions.

Universal reachability is an additional property and excludes nonconstant invariants preserved by every transition. Reliable selection requires target information in the controls even when seed entropy is zero. A singleton starting input, an empty input, and a symmetric random input have distinct mathematical meanings and distinct costs. None of these constructions removes the need to specify the underlying sets, rules, or probability laws.

References

[1] J. J. M. M. Rutten, Universal coalgebra: a theory of systems, *Theoretical Computer Science* **249** (2000), 3–80. doi:10.1016/S0304-3975(00)00056-6.

[2] C. R. Shalizi and J. P. Crutchfield, Computational mechanics: pattern and prediction, structure and simplicity, *Journal of Statistical Physics* **104** (2001), 817–879. arXiv:cond-mat/9907176.

[3] N. Barnett and J. P. Crutchfield, Computational mechanics of input–output processes: structured transformations and the ϵ-transducer, *Journal of Statistical Physics* **161** (2015), 404–451. doi:10.1007/s10955-015-1327-5.

[4] R. Givan, T. Dean, and M. Greig, Equivalence notions and model minimization in Markov decision processes, *Artificial Intelligence* **147** (2003), 163–223. doi:10.1016/S0004-3702(02)00376-4.

[5] The Stacks Project Authors, Initial and final objects, Tag 002B. `https://stacks.math.columbia.edu/tag/002B`.

[6] *Contraction, Atomic Capacity, and Exact One-Seed Simulation*, companion mathematical note (2026), `closure_atomic_capacity_one_seed.tex`.